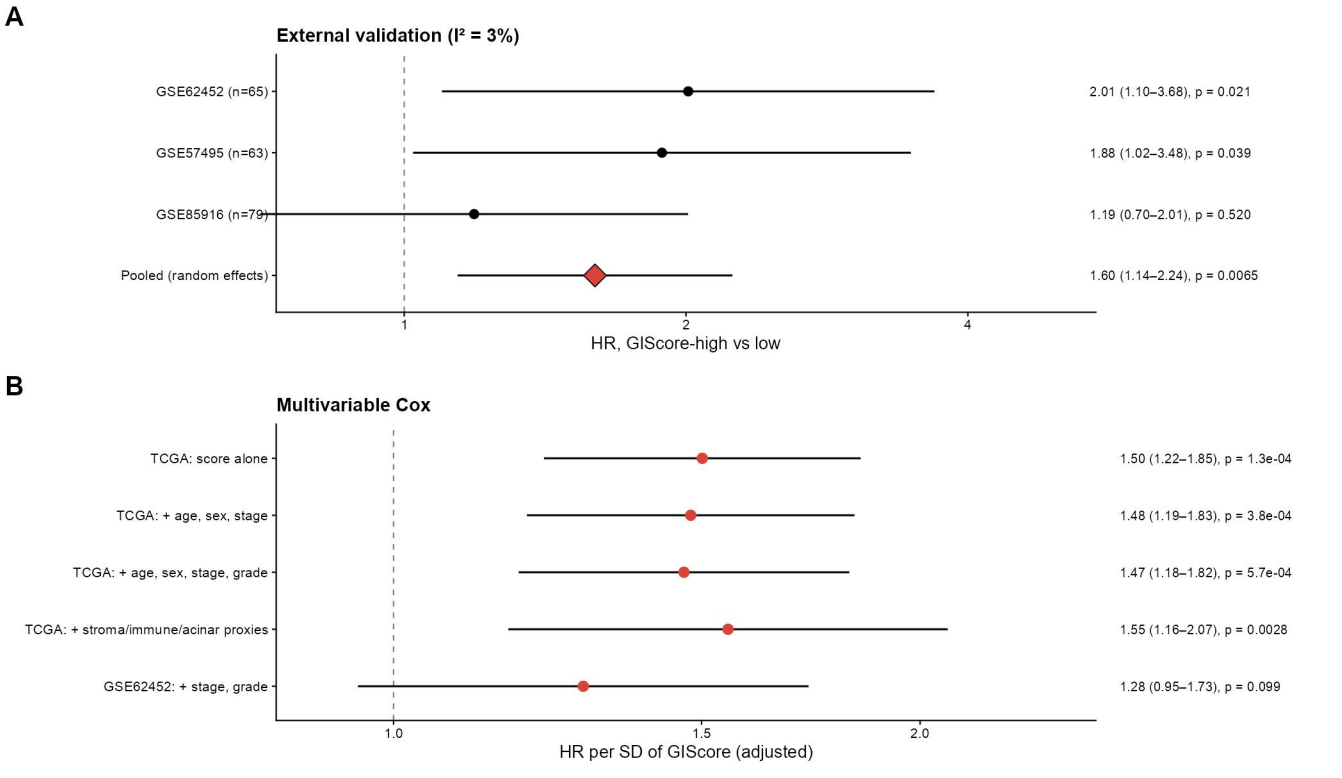
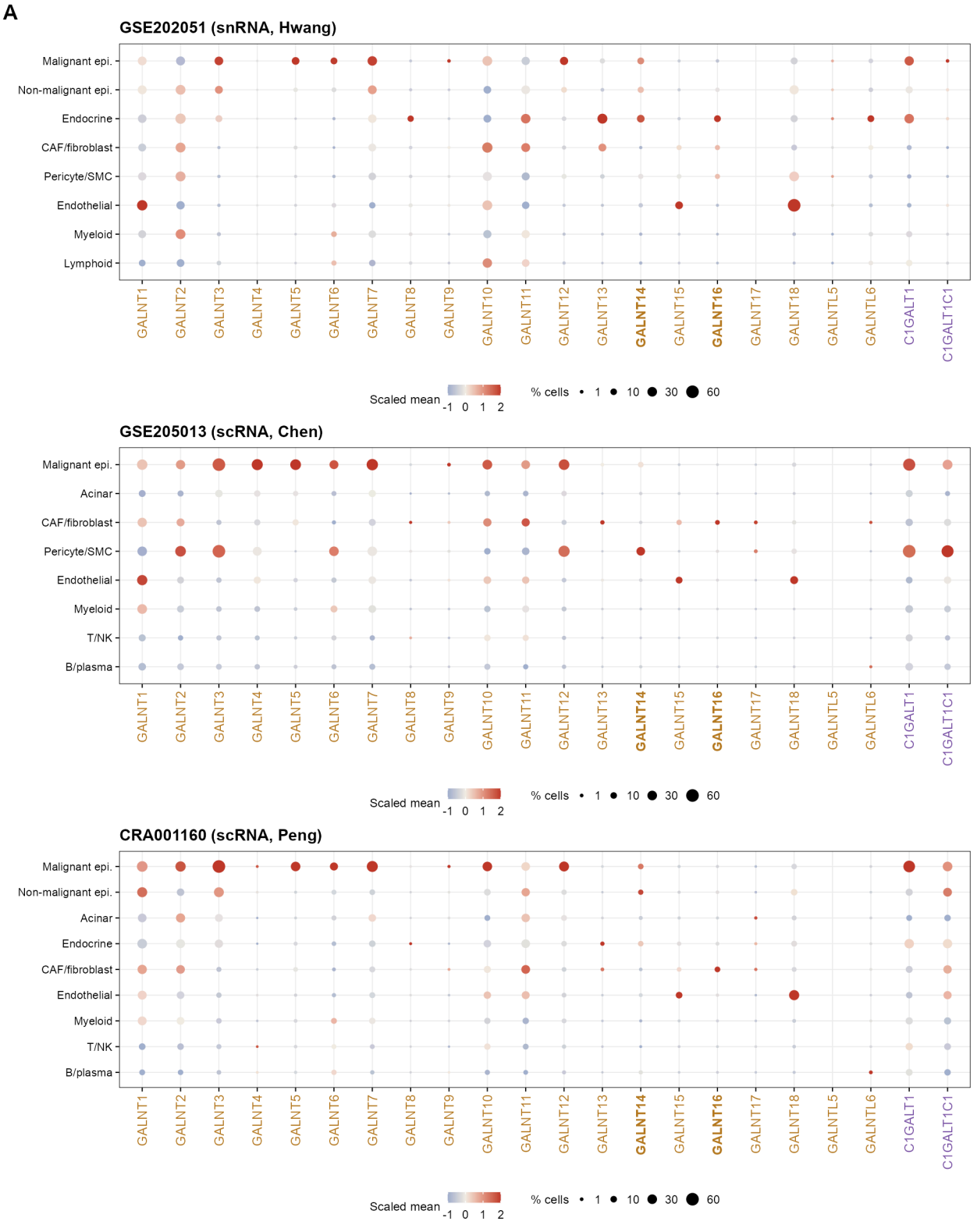
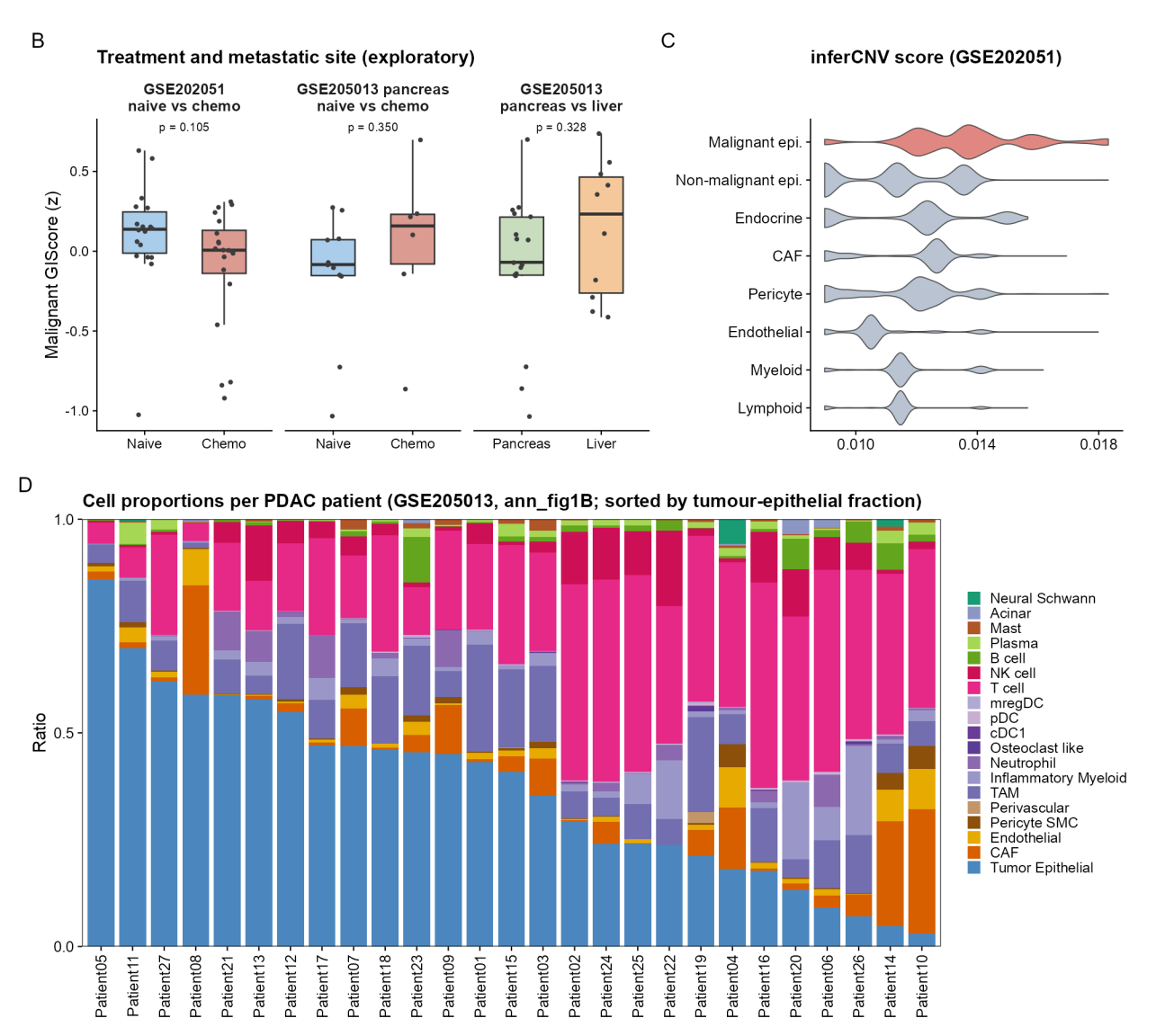
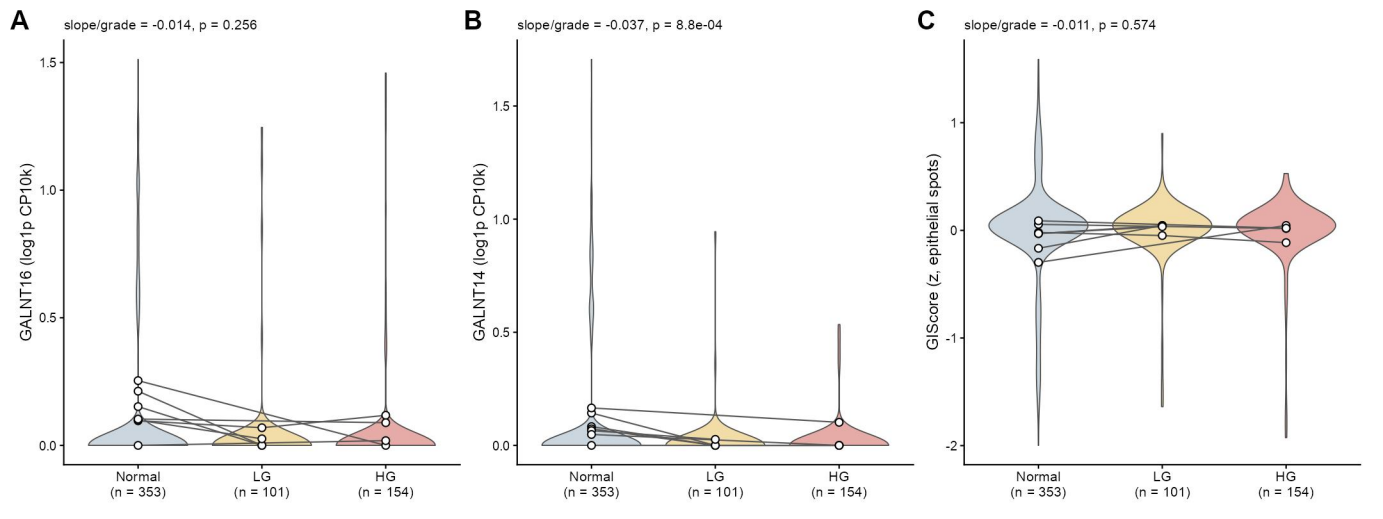
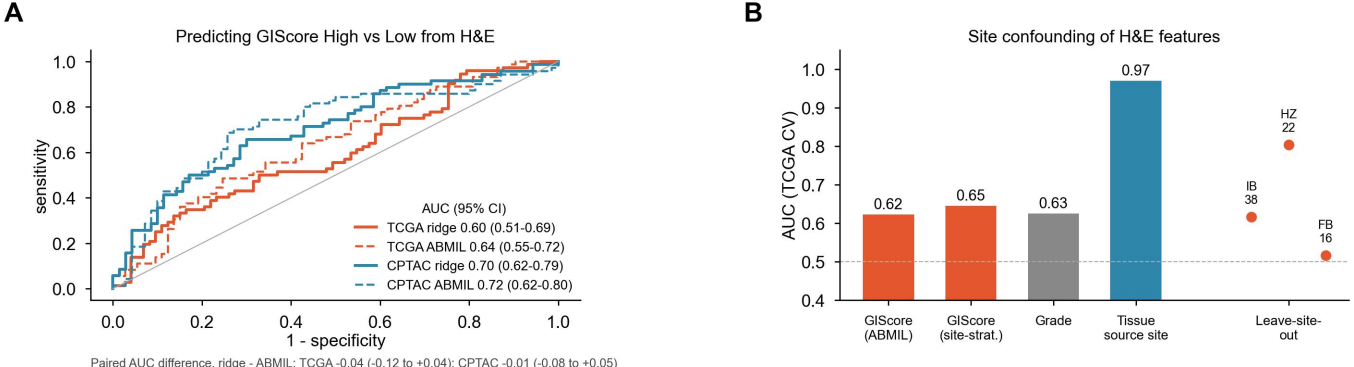

A
External validation (I² = 3%)
GSE62452 (n=65)
GSE57495 (n=63)
GSE85916 (n=79)
Pooled (random effects)
2.01 (1.10–3.68), p = 0.021
1.88 (1.02–3.48), p = 0.039
1.19 (0.70–2.01), p = 0.520
1.60 (1.14–2.24), p = 0.0065
1
2
4
HR, GIScore-high vs low
B
Multivariable Cox
TCGA: score alone
TCGA: + age, sex, stage
TCGA: + age, sex, stage, grade
TCGA: + stroma/immune/acinar proxies
GSE62452: + stage, grade
1.50 (1.22–1.85), p = 1.3e-04
1.48 (1.19–1.83), p = 3.8e-04
1.47 (1.18–1.82), p = 5.7e-04
1.55 (1.16–2.07), p = 0.0028
1.28 (0.95–1.73), p = 0.099
1.0
1.5
2.0
HR per SD of GIScore (adjusted)


A
GSE202051 (snRNA, Hwang)
Malignant epi.
Non-malignant epi.
Endocrine
CAF/fibroblast
Pericyte/SMC
Endothelial
Myeloid
Lymphoid
GALNT1
GALNT2
GALNT3
GALNT4
GALNT5
GALNT6
GALNT7
GALNT8
GALNT9
GALNT10
GALNT11
GALNT12
GALNT13
GALNT14
GALNT15
GALNT16
GALNT17
GALNT18
GALNTL5
GALNTL6
C1GALT1
C1GALT1C1
Scaled mean -1 0 1 2
% cells 1 10 30 60
GSE205013 (scRNA, Chen)
Malignant epi.
Acinar
CAF/fibroblast
Pericyte/SMC
Endothelial
Myeloid
T/NK
B/plasma
GALNT1
GALNT2
GALNT3
GALNT4
GALNT5
GALNT6
GALNT7
GALNT8
GALNT9
GALNT10
GALNT11
GALNT12
GALNT13
GALNT14
GALNT15
GALNT16
GALNT17
GALNT18
GALNTL5
GALNTL6
C1GALT1
C1GALT1C1
Scaled mean -1 0 1 2
% cells 1 10 30 60
CRA001160 (scRNA, Peng)
Malignant epi.
Non-malignant epi.
Acinar
Endocrine
CAF/fibroblast
Endothelial
Myeloid
T/NK
B/plasma
GALNT1
GALNT2
GALNT3
GALNT4
GALNT5
GALNT6
GALNT7
GALNT8
GALNT9
GALNT10
GALNT11
GALNT12
GALNT13
GALNT14
GALNT15
GALNT16
GALNT17
GALNT18
GALNTL5
GALNTL6
C1GALT1
C1GALT1C1
Scaled mean -1 0 1 2
% cells 1 10 30 60


B
Treatment and metastatic site (exploratory)
GSE202051 naive vs chemo
GSE205013 pancreas naive vs chemo
GSE205013 pancreas vs liver
p = 0.105
p = 0.350
p = 0.328
Malignant GIScore (z)
0.5
0.0
-0.5
-1.0
Naive
Chemo
Naive
Chemo
Pancreas
Liver
C
inferCNV score (GSE202051)
Malignant epi.
Non-malignant epi.
Endocrine
CAF
Pericyte
Endothelial
Myeloid
Lymphoid
0.010
0.014
0.018
D
Cell proportions per PDAC patient (GSE205013, ann_fig1B; sorted by tumour-epithelial fraction)
Ratio
1.0
0.5
0.0
Patient05
Patient11
Patient27
Patient08
Patient21
Patient13
Patient12
Patient17
Patient07
Patient18
Patient23
Patient09
Patient01
Patient15
Patient03
Patient02
Patient24
Patient25
Patient22
Patient19
Patient04
Patient16
Patient20
Patient06
Patient26
Patient14
Patient10
Neural Schwann
Acinar
Mast
Plasma
B cell
NK cell
T cell
mregDC
pDC
cDC1
Osteoclast like
Neutrophil
Inflammatory Myeloid
TAM
Perivascular
Pericyte SMC
Endothelial
CAF
Tumor Epithelial


A
slope/grade = -0.014, p = 0.256
GALNT16 (log1p CP10k)
1.5
1.0
0.5
0.0
Normal
(n = 353)
LG
(n = 101)
HG
(n = 154)
B
slope/grade = -0.037, p = 8.8e-04
GALNT14 (log1p CP10k)
1.5
1.0
0.5
0.0
Normal
(n = 353)
LG
(n = 101)
HG
(n = 154)
C
slope/grade = -0.011, p = 0.574
GIScore (z, epithelial spots)
1
0
-1
-2
Normal
(n = 353)
LG
(n = 101)
HG
(n = 154)


A
Predicting GIScore High vs Low from H&E
sensitivity
1 - specificity
AUC (95% CI)
TCGA ridge 0.60 (0.51-0.69)
TCGA ABMIL 0.64 (0.55-0.72)
CPTAC ridge 0.70 (0.62-0.79)
CPTAC ABMIL 0.72 (0.62-0.80)
Paired AUC difference, ridge - ABMIL: TCGA -0.04 (-0.12 to +0.04); CPTAC -0.01 (-0.08 to +0.05)
ABMIL pre-specified; ridge post hoc (nested alpha). CPTAC: models trained on all TCGA slides.
B
Site confounding of H&E features
AUC (TCGA CV)
0.62
0.65
0.63
0.97
HZ
22
IB
38
FB
16
GIScore (ABMIL)
GIScore (site-strat.)
Grade
Tissue source site
Leave-site-out